



Multi-dimensional energy poverty and inequality in Vietnam

Nguyen Doan¹ · Canh Phuc Nguyen^{1,2} · Thanh Dinh Su¹ · Binh Quang Nguyen¹ · Duyen Thuy Le Tran³

Accepted: 16 May 2025 / Published online: 9 June 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract

Although Vietnam has improved electricity access significantly over recent decades, energy poverty remains a challenge, particularly among the disadvantaged population. Existing studies often overlook the complexity of energy poverty by focusing on single-dimensional measures. This study provides a comprehensive analysis of energy poverty across different households and regions in Vietnam between 2004 and 2020, focusing on multi-dimensional aspects and associated inequalities in energy poverty. Building on four main measures of energy poverty (10% indicator (TPR), low-income high-cost (LIHC), half-median spending (M2), and limited modern appliances (MOT)), we account for different dimensions of energy poverty, including income, expenditure, affordability, and access to modern technologies. A multi-dimensional energy poverty index (MEPI) measure computed by different statistical methods consistently reveals that energy poverty is mainly concentrated across districts in regions with low economic development and underdeveloped infrastructure. By leveraging household data and using the coefficient of variation, we also identify significant gaps in energy poverty levels within household socio-economic and demographic characteristics. We find the groups most vulnerable to energy poverty are households with the oldest or youngest heads, those with no formal schooling, single or widowed individuals, rural residents, and those in the lowest income quintile. The findings emphasize the need for targeted policies and programs, such as subsidies for vulnerable households, investment in rural energy infrastructure, and initiatives to enhance access to modern appliances. This study contributes to the broader understanding of energy poverty by offering a novel strategy for measuring energy poverty in a developing country and calls for action to reduce energy poverty and inequality and foster equitable access to energy in a developing country.

Keywords Energy poverty · Measurement · Emerging economy · Sustainable development

JEL Classification O13 · P28 · Q41 · Q58

1 Introduction

Energy is one of the most important pillars of the United Nations Sustainable Development Goals (SDGs). Goal 7 of the 17 SDGs emphasizes that access to clean and affordable energy is key for development in all sectors.¹ Energy poverty is a “*situation in which a person or household is unable to attain energy services at a socially and materially necessary level*” (Simcock 2020, p.1) and is recognized as a crucial issue in sustainable development (Guevara et al. 2023).

According to Al Kez et al. (2024), energy poverty in developing countries relates to the lack of access to basic clean energy and the affordability of energy. There is debate about the use of a single indicator or a multi-dimensional indicator of energy poverty (Sy and Mokaddem 2022). Sy and Mokaddem (2022) indicate that scarcity of data is one of the main challenges in measuring energy poverty in developing countries. Scholars have endeavored to address this issue by finding ways of measuring energy poverty. Awan et al. (2013) measure energy poverty in Pakistan using a multi-dimensional approach over the period 2007–2008 and find a high percentage of energy poverty in rural areas. Zhang et al. (2019) measure energy poverty in China using a similar approach and suggest that energy poverty has a significant impact on health. Studying Indonesia, Rizal et al. (2024) apply a multi-dimensional approach and conclude that energy poverty varies by education, gender, employment status, and business field of households, while it also varies across regions.

This study contributes to this line of the literature by measuring energy poverty in Vietnam over the period 2004–2020. Energy poverty is a global problem, even though it is raised mainly in the context of developing countries, where it is of concern to governments and political parties (Castaño-Rosa et al. 2019b). Importantly, energy poverty has several socio-economic and environmental implications (Lu and Ren 2023), such as education (Apergis et al. 2022), health (Omar and Hasanujjaman 2021), gender inequality (Nguyen and Su 2021b), and economic development (Amin et al. 2020). However, energy poverty is difficult to quantify and monitor (Al Kez et al. 2024), and several scholars have addressed the measurement of energy poverty (e.g., Herrero (2017), Siksnyte-Butkiene et al. (2021) for recent reviews). Therefore, understanding the status and evolution of energy poverty could have significant implications for policymakers in the field of sustainable development strategy.

This study offers two innovations to the literature. It is the first reported research to measure multi-dimensional energy poverty across regions in developing countries, and also to measure the inequality of energy poverty. The measurement of energy poverty from a multi-dimensional view for developing countries is still an important topic in the literature on energy (Wang et al. 2023; Manasi and Mukhopadhyay 2024; Rizal et al. 2024). First, our study proposes a novel strategy for measuring energy poverty in a developing country from a multi-dimensional view. The literature on multi-dimensional energy poverty suggests measuring energy poverty from various dimensions, including accessibility and affordability (income or expenditure perspective). Our study proposes that energy poverty can also be considered from the perspective of lacking electric appliances. This approach aligns with the fact that accessibility is adequately addressed in most countries, while economic development helps mitigate energy poverty from the standpoint of income or expenditure in developing countries. A household may have an electricity connection,

¹ For more details on Goal 7, see: Affordable and clean energy at <https://www.un.org/sustainabledevelopment/energy/>

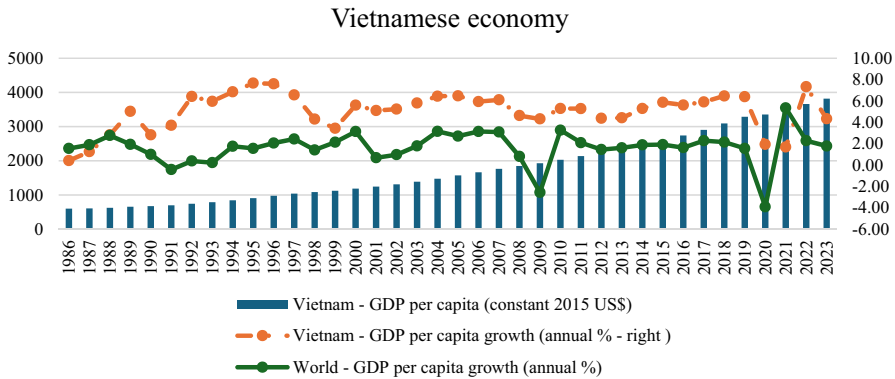


Fig. 1 Vietnam's economy over the period 1986–2023. *Note:* the data is collected from the World Development Indicators database of the World Bank (version 2024)

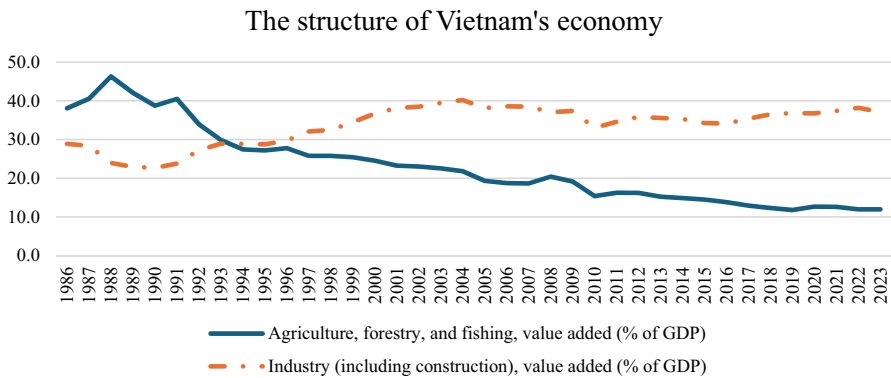


Fig. 2 The structure of Vietnam's economy over the period 1986–2023. *Note:* the data is collected from the World Development Indicators database of the World Bank (version 2024)

but without modern electrical appliances for basic needs (such as washing machines or electric fans), their living standards could be degraded. Additionally, there is a notable lack of attention to the aspect of inequality in energy poverty within the existing literature. Among the few studies, Aristondo and Onaindia (2018) show the inequality of energy poverty among population groups in Spain. This study further emphasizes that inequality of energy poverty exists not only among different population groups but also across regions.

Regarding study context, Vietnam is a special case to focus on in studying energy poverty. Vietnam has a unique economy, which was transformed in the 1980s from a centrally planned economy to a “*socialist market economy under state guidance*” (Beresford 2008). Vietnam has witnessed significant changes over the last three decades (Diez 2016), with the economy growing from low-income to lower-middle-income level² and a tripling of GDP per capita (see Fig. 1). GDP per capita growth is now much higher than the global average, and the economy has been transformed from agricultural to industrial (see Fig. 2).

² See the 2024 income classification of the World Bank at <https://blogs.worldbank.org/en/opendata/world-bank-country-classifications-by-income-level-for-2024-2025>.

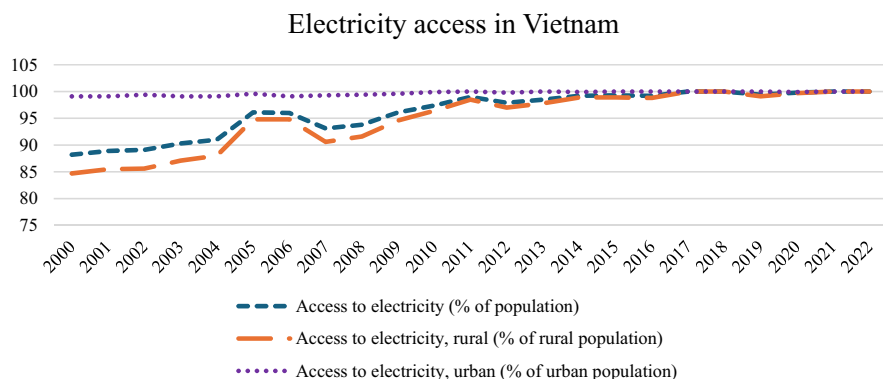


Fig. 3 Electricity access in Vietnam over the period 2000–2022. *Note:* the data is collected from the World Development Indicators database of the World Bank (version 2024)

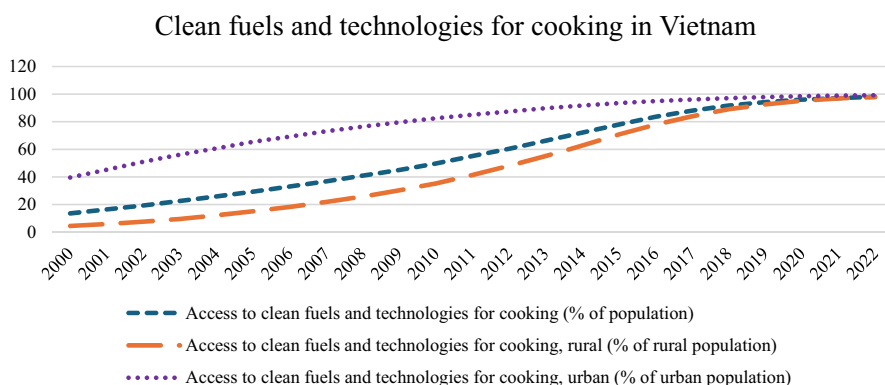


Fig. 4 Clean fuels and technologies for cooking in Vietnam over the period 2000–2022. *Notes:* the data is collected from the World Development Indicators database of the World Bank (version 2024)

Following these changes in the economic system, there has been significant progress in electricity supply, especially in urban areas (see Fig. 3). The availability of clean fuels and technologies for cooking also shows significant improvement (see Fig. 4).

According to a report by the United Nations Development Programme (UNDP), the rate of household electricity access in Vietnam increased significantly, from 48.9% to 90.5% over the period 1992–2004 (UNDP 2007). However, UNDP emphasizes that there are still significant differences between regions and groups. Nguyen et al. (2019) describe a transition in energy consumption from traditional to modern energy sources in Vietnam over the period 2004–2016, but there is still a high dependence on traditional energy sources (i.e., coal, biomass) in ethnic and rural households. The authors explain that there has been a decrease in electricity poverty, but energy-cost poverty has now emerged. Recently, Ha-Duong and Nguyen (2021) find that while electricity accessibility has increased significantly, many households (24%) still report that they cannot use electricity for their needs. These realities suggest that measuring energy poverty in a transitioning emerging economy such as Vietnam fails to provide an accurate picture if based on a simple indicator, such as access to electricity.

A combination of socio-economic and environmental factors drives energy poverty in Vietnam. Despite significant development in electricity access, especially through rural electrification programs, access alone does not ensure adequate or affordable energy use for all households (Ha-Duong and Nguyen 2021; Feeny et al. 2021). One of the main causes is low household income,³ which limits the ability to afford modern energy services and appliances, particularly among rural populations and ethnic minorities (Gencer et al. 2011). This income constraint is worse in a country that still depends on agriculture, making many households vulnerable to external shocks such as temperature extremes. Temperature shocks negatively affect agricultural productivity, reducing household income and increasing energy poverty through both lower purchasing power and higher energy needs for cooling (Feeny et al. 2021; Nguyen et al. 2019). Moreover, there are disparities not only in access but also in the volume and quality of energy, given that inequality in electricity consumption is relatively high compared to income inequality (Son and Yoon 2020). Many households continue to depend on traditional fuels like biomass and coal (Nguyen et al. 2019), indicating that energy poverty in Vietnam is not only a matter of connection to the grid but also of affordability and transition to clean energy sources (Pouran et al. 2022). Policy measures such as lifeline tariffs have improved affordability for some, but a significant share of households still report insufficient electricity consumption to meet their basic needs (Andrew and Romilly 2014). Thus, energy poverty in Vietnam is explained by a combination of factors, including household income, costs, inequality in energy use, and slow transition to modern energy technologies.

Against this background, our study employs a multi-dimensional approach to assess energy poverty. *First*, we calculate four documented measures that account for different dimensions leading to energy poverty, namely, income, expenditure, affordability, and access to modern technologies: the ten-percent index (TPR), low-income high-cost (LIHC), half-median spending (M2), and limited modern appliances (MOT). Household-level data from the Vietnam Household Living Standards Survey (VHLSS) 2004–2020 are analyzed to explore energy poverty patterns across time and areas in Vietnam. *Second*, we employ different statistical methods (unweighted averaging, principal component analysis (PCA), factor analysis, Z-Score normalization, and weighted index) to compute a composite index for multi-dimensional energy poverty. *Third*, to examine inequality in energy poverty, we calculate the coefficient of variation within different socio-economic and demographic groups of households. This analysis aims to identify vulnerable groups and shed light on gaps in energy consumption that policymakers can address.

This study is organized as follows. The next section reviews the relevant literature on energy poverty and its measurement. Section 3 outlines the data and methodology for measuring energy poverty by four indicators: TPR, LIHC, M2, and MOT. Section 4 presents the different strategies for computing a composite measure to capture the multi-dimensions of energy poverty. Section 5 develops a method to measure inequality in energy poverty across household characteristics and includes the discussion. The final section concludes with some remarks.

³ Several studies on multi-dimensional poverty, such as Solaymani et al. (2019), propose using indicators such as the headcount ratio (H) and the intensity of poverty (A) to evaluate poverty through different aspects of deprivation. Our study shares some similarities in measuring energy poverty. For example, we calculate the ratio of households considered energy poor at the district level in Vietnam. We also divide subsamples based on household characteristics, such as residence status, income quintiles, and other demographic and economic factors, and then measure inequality in energy poverty.

2 Literature review

2.1 Energy poverty: a crucial issue in sustainable development

Energy poverty is an important issue and emerging field in energy economics and is the subject of enthusiastic debate (Lu and Ren 2023; Katoch et al. 2024). According to Simcock (2020, p.1), the literature views energy poverty from two distinct perspectives: accessibility and affordability. Simcock (2020, p.1) explains that studies of energy poverty in low-income countries mostly consider energy poverty as the inability to physically access modern energy services, such as modern fuels and technologies for cooking. In other studies, mostly in European and developed countries (Campagna et al. 2024; Kashour and Jaber 2024), scholars view energy poverty from the perspective of affordability, where households have physical access to an electricity supply and other modern energy sources but cannot afford the monetary cost of consuming energy at the required level (Simcock 2020, p.1).

According to Min et al. (2024), even though access to electricity has increased significantly in recent decades, in 2020, 733 million people lacked electricity connections, and over 1.18 billion people could not use electricity or were energy-poor. Min et al. (2024) further show that 447 million people do not use electricity even though they are reported to have electricity connections in official records. Studying 146 countries from 2002 to 2018, Salman et al. (2022) conclude that energy poverty has improved in developing countries since 2008, while Africa and Central and South Asia remain regions with higher energy poverty. Several studies have been carried out to measure energy poverty in developing countries at the country level (e.g., García Ochoa and Graizbord Ed (2016) for Mexico, Olawumi Israel-Akinbo et al. (2018) for South Africa, Zhang et al. (2019) for China, and Gupta et al. (2020) for India). Jayasinghe et al. (2021) apply the multi-dimensional approach to measure energy poverty and conclude that Sri Lankan households are exposed to a moderate level of energy poverty. Wang et al. (2023) show that a high level of energy poverty exists in China, especially in rural areas.

Energy poverty is important to sustainable development because it is a worldwide societal and political challenge (see Lu and Ren (2023) for a recent review), with severe consequences on both physical and mental health, and with impairment of social, political, and economic opportunities. Guevara et al. (2023) indicate that energy poverty causes 100,000 excess deaths annually worldwide. Sovacool (2012) concludes that energy poverty can lead to indoor air pollution and physical injury, while it also exacerbates gender inequality and deforestation and increases emissions. Nguyen and Su (2021b) find that energy poverty can increase gender inequality. Nguyen and Su (2021c) add that energy poverty had a negative impact on productivity in a sample of 45 developing countries from 2002 to 2017. Energy poverty is also the subject of attention because it has strong connections with other issues, which pose further challenges to policymakers. Nguyen and Nasir (2021) find that energy poverty and income inequality are associated in a sample of 51 economies over the period 2002–2014. Nguyen and Su (2021a) suggest that energy poverty and deforestation have mutual causalities. Nguyen and Dinh Thanh (2022) document mutual causality between energy poverty and economic vulnerability in 73 low- and middle-income economies.

2.2 The measurement of energy poverty: a summary

There has been extensive discussion about the measurement of energy poverty, given its complex and multi-faceted nature (Middlemiss et al. 2019; Pérez-Fargallo et al. 2023). The absence of agreement on definitions and the lack of data are reasons why the measurement of energy poverty is still the subject of debate (Sy and Mokaddem 2022). With an early definition of energy poverty, Reddy et al. (2000, p. 44) state that energy poverty is “the absence of sufficient choice in accessing adequate, affordable, reliable, high quality, safe and environmentally benign energy services to support economic and human development.” Likewise, González-Eguino (2015) explains energy poverty as i) the absence of choice, implying no access to energy; ii) the demand for energy services, implying the provision of various energy sources; iii) adequate technologies in accessing energy services, implying affordability of energy services; iv) reliability of energy services, implying no breaks in service.

Notably, Pachauri and Spreng (2011) suggest three alternative measures of energy poverty: a technological threshold approach, a physical threshold approach, and an economic threshold approach (see González-Eguino (2015) for a summary). In the first approach, energy poverty is viewed from the perspective of accessibility, where energy poverty can be measured as a percentage of the population without access to energy services, such as electricity. In the second approach, energy poverty is viewed from the perspective of energy consumption to meet basic needs, where those with low energy consumption less than a threshold are experiencing energy poverty. In the third approach, energy poverty is viewed from the affordability perspective, where people who spend more than a reasonable percentage of their income on energy would be regarded as being in energy poverty. Each approach has its limitations (González-Eguino 2015). The first approach does not include information on the level of energy consumption, the second faces problems in identifying basic needs in energy consumption and distinguishing energy consumption from energy use in production, and the third faces the issue that the purchasing power of a currency, energy prices, and energy demand vary across countries, which leads to difficulties in cross-country comparisons.

Herrero (2017) summarizes three main approaches to measuring energy poverty. First, the direct approach defines a standard for domestic energy services (pre-defined standard) and compares the domestic energy services of households with that standard. Second, the income/expenditure approach focuses on the linkage between household income and domestic energy expenditure. Third, the consensual approach measures energy poverty based on self-assessment of a household's living conditions. The income/expenditure approach was developed and used in the EU from the 1980s, where Boardman's 10% indicator (TPR) and the low-income high-cost (LIHC) indicator are among the most commonly used methods (Herrero 2017). Boardman's 10% indicator states that if a household spends 10% or more of its income on energy, it is considered to be in energy poverty.⁴ The LIHC indicator⁵ determines energy poverty by focusing on the income-poor strata. It defines households as energy poor, where a household is considered as experiencing energy poverty if its remaining income after spending on energy services is lower than the official poverty threshold, while their spending on energy services is higher than the median energy cost for the population (Siksnyte-Butkiene et al. 2021). Other income/expenditure indicators have been developed, such as the

⁴ This indicator originates from Boardman (1991)

⁵ This indicator originates from Hills (2012)

twice-the-national-median indicator, the minimum income standard indicator, or the after-fuel-cost poverty indicator (Siksnylyte-Butkiene et al. 2021). With the twice-the-national-median indicator, energy poverty exists if a household spends double the median on energy services (Castaño-Rosa et al. 2019a). Under the minimum income standard indicator, energy poverty exists if a household's income after energy services spending is lower than the minimum income standard (Moore 2012). With the after-fuel-cost poverty indicator, energy poverty exists if a household cannot pay for essential energy services after spending on housing and other basic needs (Hills 2011).

The literature also considers the choice between single and multi-dimensional indicators in measuring energy poverty. The percentage of the population with an electricity connection and non-solid fuel consumption for cooking are still popular single indicators (Castaño-Rosa et al. 2019b), but Herrero (2017) indicates that using a single indicator is problematic since it does not facilitate a comprehensive understanding of the phenomenon of energy poverty. Multi-dimensional indicators are used more frequently, though they face problems with definitions of energy poverty, and there is debate about the dimensions that should be used in measurement (Castaño-Rosa et al. 2019b). Indicators that have emerged include the World Bank's multi-tier framework for measuring energy access (Bhatia and Angelou 2015), the total energy access approach (Action 2013), and the multi-dimensional energy poverty index (Nussbaumer et al. 2012, 2013). These indicators assume that basic energy services for human well-being include energy for cooking, lighting, entertainment and communications, space cooling and heating, refrigeration, mechanical loads, and product heating (Bhatia and Angelou 2014).

Carrying out a systematic review, Siksnylyte-Butkiene et al. (2021) find that previous studies use various indicators to measure energy poverty in different contexts. Single indicators such as the Boardman 10% indicator or LIHC are popular (e.g., Legendre and Ricci (2015), März (2018), Galvin (2024)) since they can be adapted according to national objectives and standards (Siksnylyte-Butkiene et al. 2021). Meanwhile, the multi-dimensional energy poverty index based on different dimensions is also used (e.g., Olang et al. (2018), Ahmed and Gasparatos (2020), Rizal et al. (2024)). Siksnylyte-Butkiene et al. (2021) conclude that energy poverty is a complex issue that depends on various factors and that simple indicators or multi-dimensional indicators have different advantages and disadvantages. Sy and Mokaddem (2022) describe three approaches for measuring energy poverty in developing countries: i) single indicators, including an engineering-based method using the minimum energy requirement for basic energy services and an economics-based method using income/expenditure; ii) dashboards with a set of non-aggregated indicators to provide a broad picture of a country's energy system along with progress in achieving the SDGs; and iii) composite indices using multi-dimensional indicators. Sy and Mokaddem (2022) conclude that each approach has advantages and disadvantages and that, while the multi-dimensional approach is emerging as the preferred approach, data scarcity may limit its popularity.

3 3. Measuring energy poverty in Vietnam 2004–2020

3.1 Data and sample

To measure energy poverty at the household level, this study uses the VHLSS, which contains substantial household socio-economic and demographic data. Over the period 2004–2020, VHLSS was conducted as a biannual cross-sectional survey collecting comprehensive household and individual data across all 63 provinces and 705 districts, allowing district-level

analysis of energy poverty over time.⁶ The VHLSS data was collected by the General Statistics Office of Vietnam (GSO) using a three-stage sampling process. First, around 3000 communes or wards were selected and divided into enumeration areas based on the sampling process of the 1999 or 2009 censuses. Second, three enumeration areas were randomly chosen from each commune or ward. Last, household samples were drawn from each enumeration area. The VHLSS was designed as a rolling panel with around 50% of households surveyed in one round (e.g., VHLSS 2010) also included in the next round (e.g., VHLSS 2012). Constructing a complete household panel dataset was not practical, as this would limit the sample over a long period. For this study, cross-sectional data with a good representative sample is more appropriate, and we can aggregate the data at the district level to evaluate the spatial distribution of energy poverty across regions. The sample comprises around 9,000 households surveyed across nine waves from 2004 to 2020. To measure energy poverty, we limit the sample to households with information on electricity spending, income, expenditure, and number of electrical appliances; this results in a sample of 81,146 observations spread across 8,254 wards or communes with 18,459 enumeration areas. Information on the socio-demographic and living conditions of households is collected for analysis. Appendix Table 3 provides summary statistics of all the variables in this study.

3.2 Dimensions of energy poverty

Compared to other energy sources, electricity is more efficient for basic needs, such as lighting. Thus, in this study, we follow the perspective of Lin and Wang (2020) and Sadath and Acharya (2017) to measure energy poverty by household electricity consumption. In this context, energy poverty is defined as when a household must spend an excessive amount on electricity bills. Substantial literature on this topic proposes measuring energy poverty based on household income, expenditure, or electricity quantity-based thresholds (Sadath and Acharya 2017; Feeny et al. 2021; Liddell et al. 2012; Coelho and Goldemberg 2013).

Based on this diverse perspective, we calculate different indicators to evaluate energy poverty from different economic dimensions of income, expenditure, and specific needs for electricity spending. After evaluating these component indices, we measure multi-dimensional energy poverty using survey data across countries (e.g., for India (Sadath and Acharya 2017), Vietnam (Feeny et al. 2021), and Japan (Okushima 2017)). All the energy poverty indicators in this study are binary, showing whether or not a household is considered as energy-poor.

First, TPR identifies a household as energy-poor if its electricity expense exceeds 10% of its total living costs and its total household income falls below the median income of its cohort. Hence, $TPR = 1$ is determined based on the two criteria:

$$\text{Electricity spending} > 10\% \text{ total housing costs} \quad (1)$$

$$\text{Household income} < \text{Median national household income} \quad (2)$$

⁶ There are 63 provinces and 705 districts in Vietnam that we can identify and match with VHLSS 2020 data. It is worth noting that there is inconsistency between the VHLSS and the Global Administrative Areas (GADM) datasets in the names of district-level administrative units. Over time, Vietnam has experienced splits, re-establishments, and mergers of administrative units, while the GADM data has not been kept up-to-date. To consistently define district codes in Vietnam in the period 2004–2020, we have followed the approach and data provided by McCaig et al. (2024).

Second, another measure of energy poverty, the LIHC index, is also based on household income and expenditure. It classifies households as energy-poor if their energy costs exceed the national median and their residual income is below the poverty line or 60% of the national median income. $LIHC = 1$ is thus based on two criteria, as follows:

$$\text{Electricity spending} > \text{Median national electricity spending} \quad (3)$$

Household net income

$$\begin{aligned} &< 60\% \text{ Median of national household net income with Household net income} \\ &= \text{Disposable income} - \text{Housing costs} \end{aligned} \quad (4)$$

Third, the quantity-based approach to measuring electricity poverty, as explained in Moss et al. (2020), quantifies a threshold of 100 kWh for urban households and 50 kWh for rural households in a household with five members on average. Due to inconsistent data on household electricity consumption in the VHLSS from 2004 to 2020, an alternative approach is to impute this index based on electricity spending because progressive tariff rates on retail electricity were applied at the same rate across provinces in Vietnam. Hence, this index (M2), derived from median electricity spending, classifies a household as energy-poor if its electricity expenditure is less than half the national median household electricity expenditure:

$$M2 = 1 | \text{Electricity spending} < 50\% \text{ Median national electricity spending} \quad (5)$$

These energy poverty indicators are straightforward to calculate across common household datasets; however, each may have limitations. There are potential measurement errors in self-reported income and expenditure. Importantly, these measures may misclassify wealthy households with high electricity consumption as energy-poor, while failing to account for the energy efficiency of modern techniques and electricity appliances, an important determinant of energy costs. Using residual income instead of gross income, the LIHC index might be more effective than TPR in identifying low-income households with high energy needs. However, the computation still depends on arbitrary thresholds and may not be able to account for many unobserved factors to explain spending behaviors at the household level. Similarly, for the M2 index, the dependence on the threshold context is complicated and inaccurate if household allocation behaviors for living costs vary significantly between communes or wards in Vietnam. Compared to LIHC and TPR, while focusing on the quantity of energy consumed is easy to interpret for the general population, the M2 index does not account for the proportion of income spent to achieve the required energy consumption, potentially overlooking households that meet the consumption threshold by dedicating a disproportionately large share of their income to energy spending.

Figure 5 depicts the spatial distribution of average household electricity spending across districts in Vietnam using data from the VHLSS 2004–2020. There is significant geographic heterogeneity in household electricity spending. Districts with higher spending are concentrated in regions such as the Red River Delta, the Southeast, or the Mekong River Delta. These regions include urban centers and the economically developed areas surrounding Ho Chi Minh City and Hanoi, which often have better access to electricity infrastructure and higher household income. In contrast, lower electricity spending is in regions with limited access to electricity, lower incomes, or rural areas with less reliance on energy-intensive consumption. This spatial representation highlights regions where energy poverty may be more pronounced. Low spending on electricity could indicate

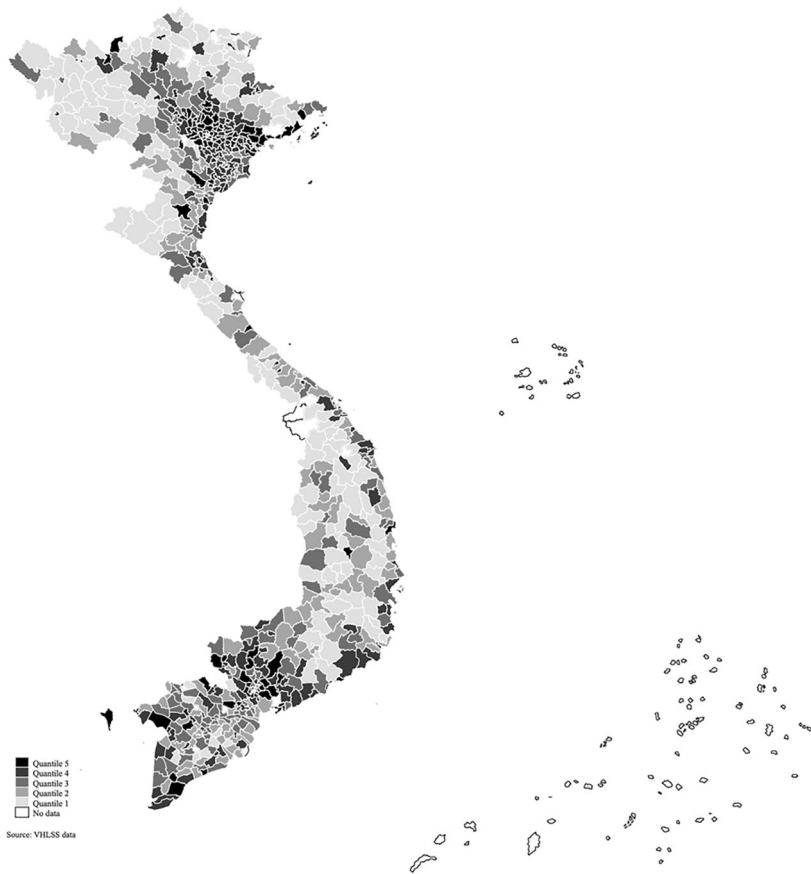


Fig. 5 Household spending on electricity in the last 12 months by quintiles. Notes: This figure illustrates the distribution of average household spending for electricity across districts in Vietnam by quintile. Data sourced from the VHLSS 2004–2020 database. Darker shades represent districts with higher electricity spending

whether energy poverty comes from limited access, affordability issues, low income, or insufficient infrastructure and resources. These observations are consistent with previous studies on energy poverty in Vietnam (e.g., UNDP (2007), Kooijman-van Dijk and Clancy (2010), Son and Yoon (2020)). That is, the Vietnamese government's electrification program has mostly eliminated energy poverty across the country in terms of the inability to access electricity (Kooijman-van Dijk and Clancy 2010). However, there are still imbalances in the use of electricity across regions (Nguyen 2022; Minh and Son 2021). The observations in our study further highlight that a majority of the Vietnamese population still spends small amounts on electricity consumption, as found in Ha-Duong and Nguyen (2021). This observation is also consistent with the conclusion in Nguyen et al. (2019) that households in low-development areas still make heavy use of traditional energy sources and shows that energy poverty from the perspective of affordability in Vietnam is still a critical issue (Nguyen et al. 2019).

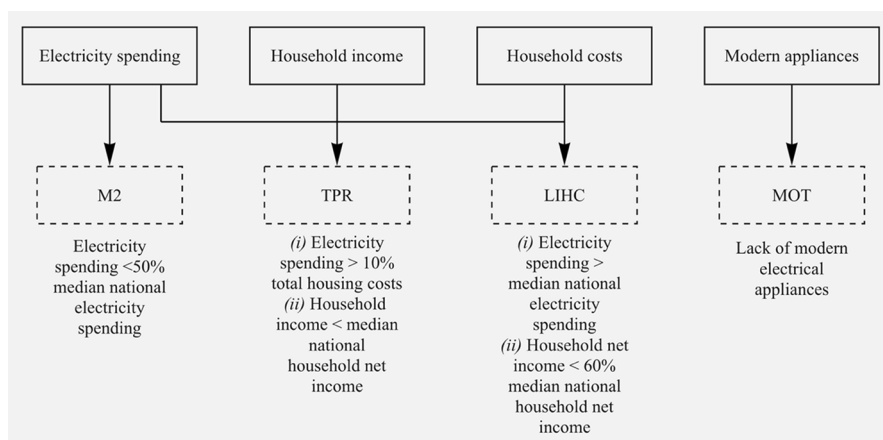


Fig. 6 The linkages among indicators of multidimensional energy poverty

Given the paucity of data, in this study, we follow a more modern approach that focuses on the quality of energy consumption instead of financial constraints on energy consumption. This approach is well-established in previous studies (e.g., Nussbaumer et al. (2012), Feeny et al. (2021)). Following these studies, we measure energy poverty by the type and quantity of modern electrical appliances used by households. We aim to implement a direct measure of how the quality of energy consumption contributes to improved living standards. Appendix Table 4 categorizes household electrical equipment into two distinct groups: essential and non-essential. Essential electrical equipment includes items fundamental to basic living standards and household functionality, such as pumps, sewing machines, televisions, refrigerators, electric cookers, and electric fans. These items are crucial for meeting basic living needs. They are assumed to be less sensitive to electricity consumption and financial conditions and are less likely to be determined by energy poverty status. In contrast, non-essential electrical equipment improves convenience and quality of life but is not critical to meeting basic living needs. This equipment includes luxury or specialized items, such as air conditioners, washing machines, vacuum cleaners, computers, and musical instruments. Households that lack non-essential but modern electrical equipment may face energy poverty, as household conditions constrain their ability to meet fundamental needs. We define a household as energy-poor if it does not have access to modern electrical appliances (MOT).

$$\text{MOT} = 1 | \text{Lack of modern electrical appliances} \quad (6)$$

Also, the distinction between essential and non-essential equipment highlights the disparity in energy access and utilization across households, implying inequalities not only in energy access and affordability but also in living standards. Thus, many households have access to electricity but still face energy deprivation and receive only limited benefits from their electricity consumption.

Figure 6 below provides an overview of the linkages between the variables, electricity spending, household income, household costs, and the availability of modern electrical appliances in constructing energy poverty indicators. Each indicator captures a specific aspect of energy poverty. In particular, the M2 index concentrates on households with exceptionally low levels of electricity spending. TPR accounts for affordability among

households where electricity spending exceeds 10% of total housing costs, conditional on income levels below the national median. Similarly, the LIHC indicator uses both electricity expenditure and household income to identify households facing a high energy cost burden with insufficient financial resources. Meanwhile, MOT measures the quality of access to electricity services by identifying households without modern electrical appliances.

In Fig. 7, we present the annual mean of energy poverty from 2004 to 2020 using four main measures: TPR, M2, LIHC, and MOT. A solid blue line represents TPR and shows a steady upward trend, indicating a consistent increase in the proportion of housing costs spent on electricity, which could be driven by increased electricity tariffs and prices or higher consumption. The validity of our approach of measuring energy poverty over a relatively long period is that we can account for the impact of economic stagnation where declines in median household income are significant, e.g., the financial crisis in 2007–2008 or the COVID-19 pandemic. The red dashed line for M2 is relatively stable over the period, indicating homogeneity in energy consumption growth that maintains the existing distribution relative to the median. This pattern emphasizes the importance of the next section in identifying potential energy inequality across different socio-economic groups to identify the most vulnerable groups. The LIHC index is indicated by the orange dashed line and exhibits a slight increase, reflecting its lower sensitivity to changes in energy affordability and household costs. However, a growing proportion of households experience high energy costs relative to the national median and low residual income after energy expenditure. An increase in household electricity spending compared to the national median may be due to inflation, higher energy tariffs, greater energy use, or even a failure of household incomes to keep pace with rising costs among vulnerable groups. Last, MOT is indicated by the purple dashed line and exhibits significant fluctuations, with a peak during the financial crisis, followed by a substantial reduction during the recent period of economic recovery

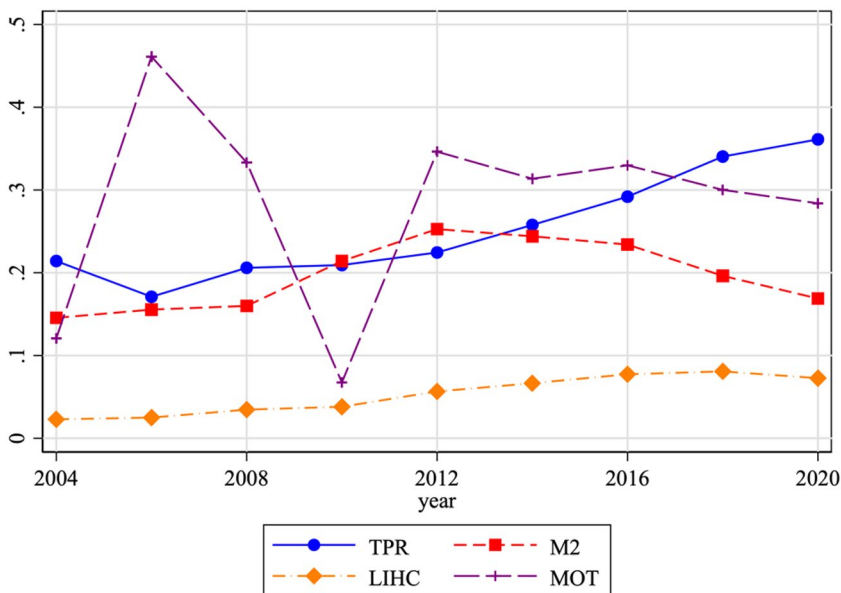


Fig. 7 Annual mean trends of energy poverty 2004–2020. Notes: This figure depicts the annual mean trends of energy poverty from 2004 to 2020 based on four measures: TPR, M2, LIHC, and MOT for different dimensions of energy poverty

and stability.⁷ Overall, it appears that the TPR and LIHC indicators of energy poverty (income/spending perspective) show a relative rise in the period 2004–2020, which reflects an increase in energy poverty in Vietnam in terms of income burden. Again, this highlights that the burden of energy consumption on income is still critical in Vietnamese households (Son and Yoon 2020). This finding is consistent with previous studies (e.g., Nguyen et al. (2019)), where the energy burden might exacerbate poverty, as the proportion of energy spending in household income might increase. Figure 7 reveals a notable trend indicating that the MOT indicator of energy poverty (electrical appliances) has been decreasing, particularly in recent years. This change may suggest an improvement in the living standards of Vietnamese households regarding modern appliances and is a positive development, as it indicates that more households in Vietnam are now able to purchase and utilize modern electrical appliances to meet their basic living needs. This improvement might be the result of economic development (Malesky 2022; Maruejols et al. 2022) and fast urbanization (Ha et al. 2021) over recent decades in Vietnam. This finding is consistent with observations in previous studies that electrical appliances are an increasing feature in household consumption, even in rural areas (e.g., Le and Pitts (2019), Nguyen and Nguyen-Ngoc (2025)).

4 A multi-dimensional energy poverty index

According to many definitions provided by different institutions about energy poverty, as discussed above, energy poverty involves a significant disruption to the functioning of a household. The situation is caused by reduced energy demand because of low income, living cost pressures, electricity costs, or a lack of access to modern technologies. This disruption can result in human, material, economic, and well-being losses, with effects that may be immediate and localized for a household or widespread and prolonged for a community. Unidimensional indicators might show different aspects of household energy poverty based on the availability, accessibility, affordability, consumption, income, living costs, or use of modern energy services. However, it is necessary to compute a composite index for energy poverty that accounts for households' varied socio-economic, demographic, and geographical characteristics. Thus, a multi-dimensional analysis framework should reflect the unique characteristics of energy poverty at the household and community levels. At the household level, the multi-dimensional energy poverty index is modeled as a combination of all dimensions of income, expenditure, affordability, and quality of energy consumption. We postulate that energy poverty for a household i is a function as follows:

$$\text{Energy poverty}_i = f(\text{Income}_i, \text{Expenditure}_i, \text{Affordability}_i, \text{Quality}_i) \quad (7)$$

In this part, we employ different approaches to compute a multi-dimensional index for energy poverty (MEPI) from all unidimensional TPR, M2, LIHC, and MOT indices, including unweighted averaging, principal component analysis (PCA), factor analysis, Z-score normalization, and a weighted index approach. The value of MEPI for all these approaches is also normalized to 0–1 for better interpretation. $MEPI = 1$ when a household is identified as multi-dimensionally energy-poor across income, expenditure, affordability, and quality.

⁷ Appendix Figure 10 shows the regional variation in different energy poverty indices by quintiles. Compared to Fig. 5, energy poverty does not always correlate directly with low electricity spending. This also implies that electricity spending alone does not indicate well-being, it must be evaluated with energy poverty measures to understand affordability and access issues.

First, in the simplest approach, we assume that the contributions of different dimensions of energy poverty are equal across all households. The unweighted averaging index is computed as the arithmetic mean of the standardized values of TPR, LIHC, M2, and MOT:

$$\text{MEPI} = \left(\sum_{j=1}^4 X_j \right)^{-1} \quad (8)$$

where X_j represents the individual energy poverty index: TPR, LIHC, M2, MOT.

Second, the PCA method is employed to compute a standardized multi-dimensional index for four single energy poverty indices: TPR, LIHC, M2, and MOT. We use the first principal component as the composite index because it explains the largest portion of variation across different energy poverty indices:

$$\text{MEPI} = \left(\sum_{j=1}^4 w_j X_j \right)^{-1} \quad (9)$$

where w_j is the weight loading of the j variable derived from the eigenvector corresponding to the largest eigenvalue of the covariance matrix. X_j is the standardized value of the j indicator, including TPR, LIHC, M2, and MOT. This statistical technique is designed to reduce the dimensionality of data while keeping as much variance as possible (Ringnér 2008). This algorithm effectively reduces a large set of variables to capture the most significant variance in energy poverty indicators; however, PCA does not explicitly account for latent variables. As such, while this data-driven method is objective, the complexity of interpreting the components might limit its application.

Developing a robust composite index to measure energy poverty requires careful consideration of the methodology because each approach offers unique strengths and limitations. The third statistical method used is factor analysis (FA), which, in contrast, is rooted in the identification of latent factors that explain the observed variables (Jolliffe 2002). Using factor analysis, we assume that the observed data are influenced by a smaller set of unobserved factors that may explain affordability and energy efficiency at the household level. This approach is particularly useful in identifying specific latent constructs driving energy poverty and when different unidimensional energy poverty indices are significantly correlated. However, this method requires strict assumptions about the data structure, and thus has a common idiosyncratic decomposition and is sensitive to model misspecification (Boivin and Ng 2006). Here, MEPI is calculated as a weighted sum of the latent factors, where the weights and the factor loadings are:

$$\text{MEPI} = \left(\sum_{j=1}^4 \lambda_j F_j \right)^{-1} \quad (10)$$

where latent factors F_j represent the unobservable dimensions that explain variance across the observed indicators TPR, LIHC, M2, and MOT. So, in the first step, each observed indicator is expressed as a combination of latent factors and unique variances, and the factor loadings quantify the contribution of each latent factor to the observed indicators:

$$\text{TPR} = \lambda_{11}F_1 + \lambda_{12}F_2 + \epsilon_1$$

$$\text{LIHC} = \lambda_{21}F_1 + \lambda_{22}F_2 + \epsilon_2$$

$$\text{M2} = \lambda_{31}F_1 + \lambda_{32}F_2 + \epsilon_3$$

$$MOT = \lambda_{41}F_1 + \lambda_{42}F_2 + \epsilon_4 \quad (11)$$

Next, Z-Score normalization is a simpler approach that standardizes each variable by subtracting its mean and dividing by its standard deviation. This method is technically straightforward and ensures that all variables are measured on the same contribution. However, it lacks the statistical rigor of PCA or FA and does not account for variance or correlation among unidimensional energy poverty indices. The MEPI calculated by the Z-score approach is as follows:

$$MEPI = \left(\sum_{j=1}^4 Z_j \right)^{-1} \quad (12)$$

where $Z_j = \frac{X_j - \mu_j}{\sigma_j}$ is the Z-score normalization standardizes each indicator TPR, LIHC, M2, MOT.

In a less ad hoc weighting scheme, instead of equal weights, an alternative approach involves estimating weights for each indicator in the multi-dimensional energy poverty index through a least-squares regression algorithm, where the dependent variable is household electricity spending. These weights are calculated to reflect the relative associations between different dimensions of energy poverty and electricity spending. The OLS regression model is as follows:

$$Electricityspending_{it} = \alpha + \beta_1 TPR_{it} + \beta_2 LIHC_{it} + \beta_3 M2_{it} + \beta_4 MOT_{it} + X_{it}\Gamma + \kappa_t + \epsilon_{it} \quad (13)$$

where $Electricityspending_{it}$ is the spending for electricity in the last 12 months of household i from year t . κ_t is year-fixed effects and ϵ_{it} is the error term. X_{it} is a vector of covariates for household characteristics, including the age, gender, and education level of the household head, the number of household members, the number of electrical appliances, the living area size, and a set of dummy variables for minority status, urban residence, and rental housing status. The estimated coefficients of TPR, LIHC, M2, and MOT are incorporated into the function for computing MEPI and assigned different contributions to each dimension of energy poverty.

$$\left(\sum_{j=1}^4 \hat{\beta}_j X_j \right)^{-1} \quad (14)$$

Figure 8 plots the distributions of MEPI computed by different weighting approaches, including unweighted averaging, PCA, FA, Z-score normalization, and weighted by electricity spending. The values of unweighted averaging and the weight index are greater than those of the other methodologies, with more values clustering on the right tails. PCA and FA produce distributions that are more evenly spread across the range of scores as a result of their statistical rigor in accounting for variability in the input data. The Z-score index is smallest on average, with lower fractions at extreme values due to its normalization effect on the data compression. These patterns indicate the need to choose appropriate methodological approaches for measuring multi-dimensional energy poverty. For instance, the clustering observed in certain methods may amplify specific dimensions of energy poverty. Advanced statistical methods for combining data, like PCA and FA, may offer a more balanced representation of the multi-dimensional aspects of energy poverty. Careful consideration in selecting a weighting approach is important for interpreting results and identifying policy implications from energy poverty measurement.

Despite the variation in outcomes depending on the method, the spatial patterns of the MEPI map across districts in Vietnam, segmented into quintiles in Fig. 9, remain

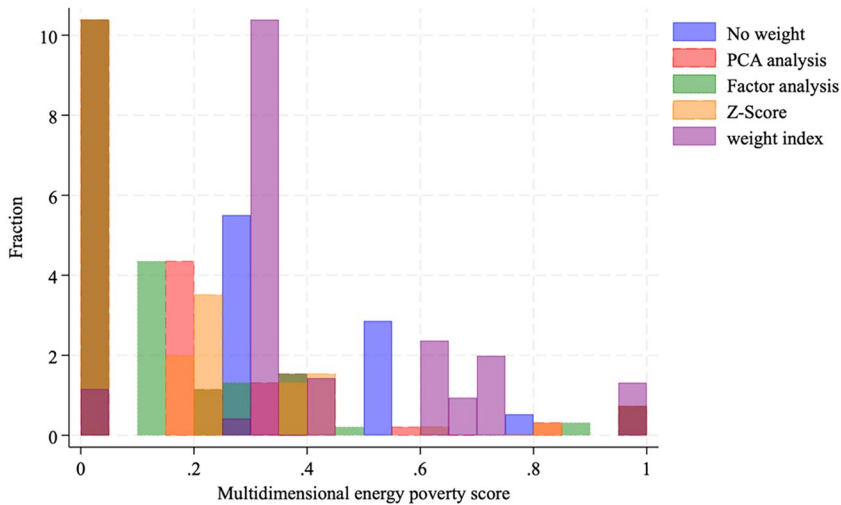


Fig. 8 Distributions of multidimensional energy poverty index. Notes: This figure shows the distributions of the multidimensional energy poverty index (MEPI) using different weighting methodologies including unweight, principal component analysis (PCA) analysis, factor analysis, Z-score, and weights by electricity spending. The x-axis represents the multidimensional energy poverty scores, while the y-axis shows the fraction of observations within each score range

broadly consistent across different approaches. The Northwest, Northeast, North Central Coast, and Central Highlands are consistently identified as the regions with the highest levels of multi-dimensional energy poverty. More specifically, the North Central Coast and Northwest regions consistently rank as the most energy-poor across all methodologies. The Central Highlands is not as pronounced in the unweighted averaging and weighted spending, but this region emerges as highly energy-deprived under PCA and FA. Conversely, the Red River Delta, Southeast, and Mekong River Delta regions, with major urban cities like Hanoi and Ho Chi Minh City, benefit from better energy infrastructure and higher economic activity and generally experience the lowest levels of energy poverty. The findings are consistent with previous studies. Regions, especially rural areas, with a low level of economic development in Vietnam appear to witness higher energy poverty (Kooijman-van Dijk and Clancy 2010; Nguyen et al. 2019; Son and Yoon 2020). This finding is also consistent with results from other studies in developing countries (e.g., Esquivel García and Toro-García (2024), Rizal et al. (2024) and others). Our calculation indicates that provinces in highland areas experience more multi-dimensional energy poverty compared to provinces near the sea. This difference might be attributed to the increased opportunities for globalization that coastal provinces have, allowing them to gain more benefits. For instance, Vo and Nguyen (2021) find that trade liberalization improves household income in Vietnam. That is, improvements in the income of households in provinces with higher exposure to trade liberalization might benefit from reduced multi-dimensional energy poverty.

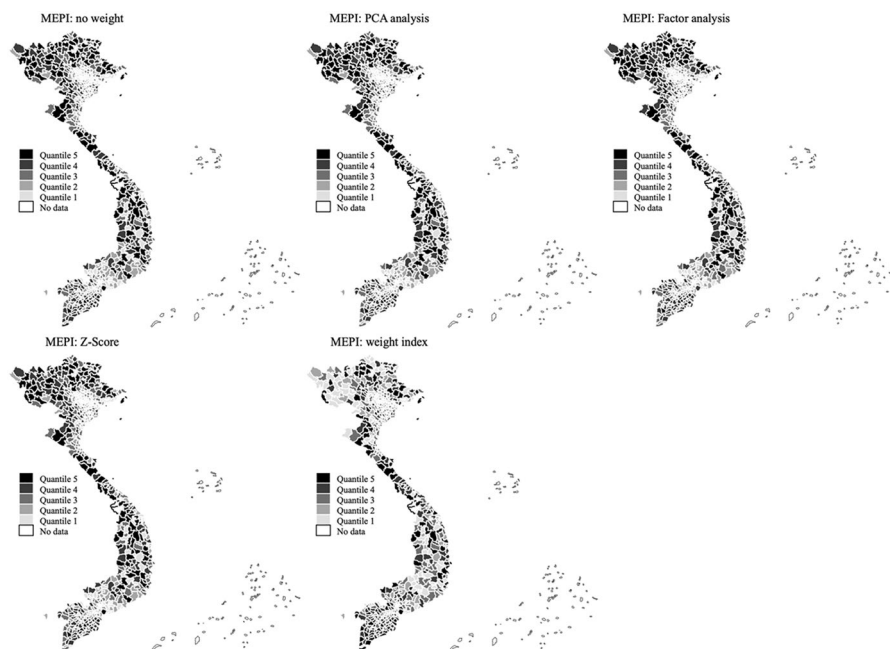


Fig. 9 Multidimensional energy poverty index by quintiles. Notes: The figure maps average energy poverty measured by MEPI using different methodologies (unweighted, PCA, FA, Z-score, and weighting by electricity spending) across districts over the period 2004–2020. Data are segmented by quintiles. The darker shades represent higher rates of energy poverty (Quintile 5), while lighter shades indicate lower rates (Quintile 1)

5 Inequality in energy poverty

The aim of this study is not only to comprehensively measure energy poverty across different households and areas in Vietnam but also to explore inequality in levels of energy poverty. Inequality in energy poverty across socio-economic and demographic groups potentially indicates significant gaps in the ability to access and afford energy and benefit from modern technology crucial to maintaining adequate living conditions. Therefore, we evaluate energy poverty across household classifications, including age, education level, marital status, household size, minority status, urban–rural divide, living space, house ownership, number of electrical appliances, and income quintiles. Each socio-economic classification is worth examining as it might be the underlying factor explaining energy poverty and its distribution within and across groups. Using four key measures, TPR, LIHC, M2, and MOT, this analysis identifies how energy poverty disproportionately affects different groups. To quantify the level of inequality, we calculate the coefficient of variation (CV) for energy poverty within each classification:

$$CV(g) = \frac{\sqrt{\frac{\sum_{i=1}^n (x_s - \mu(x))^2}{n-1}}}{\mu(x)} \times 100 \quad (15)$$

where x_s represents the average energy poverty index for each subsample s categorized by each classification g . In addition, by identifying the groups most affected by energy

poverty, the analysis might surface important policy implications. Addressing energy poverty requires a comprehensive approach considering affordability, access, infrastructure development, and energy inequality, particularly for vulnerable households.

Table 1 provides a detailed analysis of energy poverty levels across socio-economic and demographic classifications using four indices: TPR, M2, LIHC, and MOT. The findings reveal significant disparities in energy poverty levels and inequality among groups. Households with an older household head, particularly those aged 70 and above, or the youngest families with a household head under the age of 30, face the highest energy poverty levels across all indices with much higher values than the average. For example, based on TPR and MOT indices, the proportions of energy poverty among those 70 and above are 0.479 and 0.415, compared to the average of 0.251 and 0.284 of the whole population. By contrast, the age group 40–49 shows much lower values, suggesting that energy poverty may decline during people's economically productive years. Inequality within age groups is significant, particularly with LIHC of 72.55%, reflecting the disproportionate impact on the oldest or youngest households. Similarly, education classification affects inequality in energy poverty, as household heads with no schooling report the highest energy poverty levels, i.e., TPR=0.381, LIHC=0.135, while those with a high school or college education exhibit substantially lower values. Inequality linked to education level is particularly significant, with LIHC having a CV of 79.31%, which likely implies the role of education in mitigating energy poverty by enhancing income and access to energy resources. Regarding marital status, single and widowed individuals experience higher levels of energy poverty than married individuals. The results align with the gaps by the vulnerability of individuals who may lack the economic and social support often available in larger households. Household size similarly reveals that smaller households with 1–2 members have the highest energy poverty levels, whereas larger households with more than six members report much lower levels, likely due to resource pooling. Next, rural–urban disparities are also evident, with rural households consistently reporting higher levels of energy poverty. Households with rental tenure exhibit lower energy poverty levels under some measures. Last, consistent with the foundation for computing different energy poverty measures in this study, income is a critical determinant of energy poverty, with households in the lowest income quintile experiencing the highest levels of energy poverty across all indices. In contrast, the highest income quintile reports minimal energy poverty. The availability of modern energy appliances confirms the validity of our analysis, as households with fewer than three appliances face the greatest energy poverty, while those with more than 10 appliances report the lowest levels. Inequality in appliance ownership is particularly pronounced under the MOT index, with a disparity of 89.20%, while inequality within income groups is particularly severe, with LIHC showing inequality of 113.97%.

Table 2 summarizes the socio-economic and demographic groups with the highest and lowest levels of energy poverty. Panel A identifies groups with the highest energy poverty, while Panel B highlights those with the lowest. These results emphasize the significant gap in energy poverty between households. Vulnerable groups include the oldest and youngest household heads, individuals with no formal education, and minority and low-income households. The findings are consistent with several previous studies on energy poverty in developing countries. For instance, our findings are consistent with those of Manasi and Mukhopadhyay (2024) in the case of India, showing that households with higher income, higher education levels, or small household size are less likely to face energy poverty. In contrast, households in rural areas likely face the problem of energy poverty. Notably, our findings are consistent with those in previous studies in the case of Vietnam (e.g., Nguyen et al. (2019), Son and Yoon (2020), Feeny et al. (2021)).

Table 1 Energy poverty values and their inequality for different classifications

	TPR		M2		LIHC		MOT		Obs
	Coef	SE	Coef	SE	Coef	SE	Coef	SE	
Total	0.251	0.002	0.197	0.001	0.052	0.001	0.284	0.002	82,711
Age									
Under 30	0.303	0.008	0.409	0.008	0.081	0.004	0.418	0.008	3686
30–39	0.227	0.003	0.224	0.003	0.047	0.002	0.281	0.004	16,359
40–49	0.184	0.003	0.150	0.002	0.027	0.001	0.241	0.003	23,229
50–59	0.208	0.003	0.146	0.003	0.030	0.001	0.243	0.003	19,688
60–69	0.315	0.004	0.193	0.004	0.062	0.002	0.303	0.004	11,126
70 and above	0.479	0.005	0.307	0.005	0.159	0.004	0.415	0.005	8623
Inequality	37.77%	0.108	43.00%	0.102	72.55%	0.049	25.47%	0.081	
Education									
No schooling	0.381	0.005	0.366	0.005	0.135	0.003	0.415	0.005	10,123
Primary	0.264	0.003	0.220	0.003	0.055	0.002	0.319	0.003	21,298
Secondary	0.220	0.003	0.146	0.002	0.030	0.001	0.240	0.003	24,161
High school	0.183	0.004	0.073	0.002	0.013	0.001	0.148	0.003	12,069
College or above	0.254	0.004	0.234	0.003	0.062	0.002	0.325	0.004	15,060
Inequality	28.62%	0.075	52.64%	0.109	79.31%	0.047	34.71%	0.100	
Marital status									
Single	0.488	0.011	0.400	0.011	0.113	0.007	0.463	0.011	2091
Married	0.208	0.002	0.169	0.001	0.036	0.001	0.254	0.002	67,097
Widowed	0.429	0.005	0.303	0.004	0.131	0.003	0.406	0.005	11,158
Divorced	0.452	0.010	0.314	0.010	0.103	0.006	0.386	0.010	2364
Inequality	32.09%	0.127	32.18%	0.095	43.33%	0.041	23.43%	0.088	
Number of household members									
1–2	0.530	0.004	0.385	0.004	0.144	0.003	0.447	0.004	13,303
3–4	0.236	0.002	0.171	0.002	0.041	0.001	0.255	0.002	39,017
5–6	0.158	0.002	0.142	0.002	0.027	0.001	0.244	0.003	24,162
More than 6	0.122	0.004	0.173	0.005	0.026	0.002	0.267	0.006	6229
Inequality	70.82%	0.185	51.61%	0.112	95.37%	0.057	31.75%	0.096	

Table 1 (continued)

	TPR	M2		LIHC		MOT		Obs
	Coef	SE	Coef	SE	Coef	SE	Coef	SE
Total	0.251	0.002	0.197	0.001	0.052	0.001	0.284	0.002
Minority								
No	0.249	0.002	0.138	0.001	0.044	0.001	0.246	0.002
Yes	0.266	0.004	0.496	0.004	0.094	0.003	0.477	0.004
Living area								
Inequality	4.66%	0.012	79.85%	0.253	51.24%	0.035	45.18%	0.163
Rural	0.270	0.002	0.248	0.002	0.065	0.001	0.335	0.002
Urban	0.207	0.003	0.070	0.002	0.021	0.001	0.154	0.002
Inequality	18.67%	0.045	79.16%	0.126	72.35%	0.031	52.34%	0.128
Living size								
20 sq.m or less	0.449	0.010	0.396	0.010	0.139	0.007	0.513	0.010
21–40 sq.m	0.361	0.004	0.339	0.004	0.114	0.002	0.423	0.004
41–100 sq.m	0.240	0.002	0.183	0.002	0.041	0.001	0.279	0.002
More than 100 sq.m	0.140	0.003	0.058	0.002	0.009	0.001	0.113	0.003
Inequality	45.55%	0.136	62.80%	0.153	80.39%	0.061	52.69%	0.175
House ownership								
Owner	0.252	0.002	0.197	0.001	0.053	0.001	0.283	0.002
Rental	0.236	0.011	0.177	0.009	0.012	0.003	0.328	0.012
Inequality	4.63%	0.011	7.56%	0.014	89.20%	0.029	10.41%	0.032
Number of electric appliances								
3 or less	0.348	0.003	0.469	0.003	0.126	0.002	0.608	0.003
3–5	0.301	0.003	0.216	0.003	0.061	0.002	0.387	0.004
6–10	0.212	0.002	0.053	0.001	0.012	0.001	0.083	0.002
More than 10	0.083	0.003	0.005	0.001	0.001	0.000	0.001	0.000
Inequality	49.38%	0.117	112.68%	0.209	113.97%	0.057	103.81%	0.280

Table 1 (continued)

	TPR		M2		LIHC		MOT		Obs
	Coef	SE	Coef	SE	Coef	SE	Coef	SE	
Total	0.251	0.002	0.197	0.001	0.052	0.001	0.284	0.002	82,711
Income quintiles									
Quantile 1	0.417	0.004	0.471	0.004	0.173	0.003	0.466	0.004	16,233
Quantile 2	0.405	0.004	0.257	0.003	0.094	0.002	0.364	0.004	16,229
Quantile 3	0.233	0.003	0.135	0.003	N/A	N/A	0.254	0.003	16,229
Quantile 4	0.112	0.002	0.087	0.002	N/A	N/A	0.178	0.003	16,230
Quantile 5	0.045	0.002	0.054	0.002	N/A	N/A	0.110	0.002	16,225
Inequality	69.33%	0.168	84.44%	0.170	N/A	N/A	52.01%	0.143	

This table provides an overview of energy poverty using four measures (TPR, M2, LIHC, and MOT) at the household level to identify inequalities across various classifications, including age, education, marital status, number of household members, minority status, living area, living size, house ownership, number of electric appliances, and income quintiles. The coefficient and standard error of estimated household energy poverty by different classifications are reported. For each classification, the inequality values are calculated using the coefficient of variation across subsamples, highlighting the energy poverty divide within each group

Table 2 Highest and lowest energy poverty groups

	TPR	M2	LIHC	MOT
<i>Panel A. Highest energy poverty groups</i>				
Age	70 and above	Under 30	70 and above	Under 30
Education	No schooling	No schooling	No schooling	No schooling
Marital status	Single	Single	Widowed	Single
Number of household members	1–2	1–2	1–2	1–2
Minority	Minority	Minority	Minority	Minority
Living area	Rural	Rural	Rural	Rural
Living size	20 sq.m or less	20 sq.m or less	20 sq.m or less	20 sq.m or less
House ownership	Owner	Owner	Owner	Rental
Number of electric appliances	3 or less	3 or less	3 or less	3 or less
Income quintiles	Quantile 1	Quantile 1	Quantile 1	Quantile 1
<i>Panel B. Lowest energy poverty groups</i>				
	TPR	M2	LIHC	MOT
Age	40–49	50–59	40–49	40–49
Education	High school	High school	High school	High school
Marital status	Married	Married	Married	Married
Number of household members	More than 6	5–6	More than 6	5–6
Minority	No Minority	No Minority	No Minority	No Minority
Living area	Urban	Urban	Urban	Urban
Living size	More than 100 sq.m	More than 100 sq.m	More than 100 sq.m	More than 100 sq.m
House ownership	Rental	Rental	Rental	Owner
Number of electric appliances	More than 10	More than 10	More than 10	More than 10
Income quintiles	Quantile 5	Quantile 5	Quantile 5	Quantile 5

This table lists the groups with the highest and lowest levels of energy poverty from results in Table 1 across various socio-economic and demographic classifications, including age, education, marital status, number of household members, minority status, living area, living size, house ownership, number of electric appliances, and income quintiles

6 Conclusion

Energy poverty is not just a technical challenge but is also a human issue. The multi-dimensional aspects of energy poverty require more than one-size-fits-all solutions. The analysis implies that the lens through which energy poverty is measured matters, because the perspective and approach can change the focus and reveal different stories of energy poverty.⁸ Our study not only emphasizes the importance of comprehensive measures of energy poverty but also identifies how energy deprivation is faced by marginal groups, e.g.,

⁸ Appendix Figure 11 plots the cumulative distribution of all energy poverty indices analyzed in this study (TPR, M2, LIHC, MOT, and MEPI) using different methodologies.

the elderly in rural areas, single individuals, and low-income households who are living in the shadows of modern energy services.

While this study provides a novel evaluation of energy poverty in Vietnam, several limitations should be noted. First, the analysis relies on household survey data with substantial and important information about living standards. However, the cross-sectional nature of the data does not allow us to explain fully the changes in energy poverty within households over time, especially given seasonal variations in energy needs or the long-term impact of climate change or policy interventions. This limitation may lead to an underestimation of energy poverty, particularly in vulnerable areas or for households that have migrated without a permanent residence. Second, although multiple indicators are used to develop an ad-hoc multi-dimensional energy poverty index, the selection of indicators and thresholds is based on documented evidence but may be influenced by data availability and the computation process. Thus, these indicators might not capture all aspects of energy poverty, such as cultural preferences or local governance, which could also impact energy consumption patterns.

As the global energy transition accelerates and significant progress is made in meeting specific SDG-7 targets, our study shows that simple indicators of energy poverty should not be the only proxies used to construct policy measures. The findings show that despite improved electricity access in Vietnam, many households still experience energy poverty due to income, expenditure burdens, or lack of modern electrical appliances. Governments need to focus on transforming lives rather than just increasing megawatts. Additionally, multi-dimensional energy poverty varies across regions and population groups. Thus, governments should implement policies that link energy, equity, and opportunity to ensure that all households have access to necessary resources. These policies could include subsidies for vulnerable groups to access clean fuels and technologies for cooking. For instance, there is evidence that government subsidies for clean energy might be effective in promoting clean and affordable cooking in developing countries (see Tripathi et al. (2015), Astuti et al. (2019)). Our findings further advocate these policy measures. The findings of this study show that energy poverty disproportionately affects vulnerable groups like older individuals, rural residents, and low-income households. This correlation emphasizes the importance of addressing the root causes of energy poverty through policies and programs, for example subsidies, that prioritize the most disadvantaged groups. Further, this study implies that education or shared resources within larger households can act against energy poverty. Our findings support policies to invest in education systems to erase energy poverty in developing countries.

Finally, it is worth noting that governments should take appropriate measures and implement policies to tackle energy poverty. Future studies should pay more attention to this issue. For instance, Nguyen and Su (2022) suggest that government spending might have a non-linear influence on energy poverty, but the study uses accessibility to electricity/clean fuels and technologies for cooking as proxies of energy poverty for a cross-country sample. The measurement of multi-dimensional energy poverty in the present study opens new avenues for future exploration of how fiscal spending or fiscal policy could affect multi-dimensional energy poverty. More importantly, there are concerns over the consequences of climate change on energy poverty (see (Awaworyi Churchill et al. 2022), Li et al. (2023)), with some studies carried out in Vietnam (see Feeny et al. (2021), Que et al. (2022)). However, no study has explored the mechanism for these

effects. Our study has provided a comprehensive dataset of multi-dimensional energy poverty across provinces, which might offer a chance to explore this direction further. This research could lead to an examination of the effects of climate change on migration (Lukyanets et al. 2015; Trinh and Munro 2024), which then affects energy poverty (Choumert-Nkolo and le Roux 2024).

Appendix

See Tables 3, 4 and Figs. 10, 11.

Table 3 Summary statistics

	Mean	SD	Min	Max
Electricity spending in last 12 months (thousand USD)	0.084	0.114	0	11.230
Income per capita (thousand USD)	1.863	2.879	0	1140.887
Number of electrical machines	6.187	4.134	0	84
TPR: Ten-percent measure	0.252	0.434	0	1
M2: Half median measure	0.197	0.398	0	1
LIHC: Low-income high-cost measure	0.053	0.223	0	1
MOT: Limited modern appliances measure	0.284	0.451	0	1
MEPI: no weight	0.196	0.256	0	1
MEPI: PCA analysis	0.157	0.233	0	1
MEPI: Factor analysis	0.141	0.228	0	1
MEPI: Z-Score	0.169	0.238	0	1
MEPI: weight index	0.455	0.224	0	1
Age of household head	50.162	13.930	11	105
Gender of household head (1 = Male, 0 = Female)	0.752	0.432	0	1
High school graduation of household head [1 = yes, 0 = No]	0.328	0.469	0	1
Number of household members	4.123	1.655	1	20
Minority or not [1 = yes, 0 = No]	0.165	0.371	0	1
Urban area or not [1 = yes, 0 = No]	0.285	0.451	0	1
Living size (square meter)	75.852	49.193	4	2010
Rental or not [1 = yes, 0 = No]	0.020	0.139	0	1
Observations	82,711			

This table reports the summary characteristics of the variables analyzed in this study. We collect household-level data from the Vietnam Household Living Standards Survey (VHLSS) for the years 2004–2020. There are measures related to energy poverty including TPR, M2, LIHC, MOT indices based on spending and income. The multidimensional energy poverty indices MEPI calculated using different methodologies including no weight, PCA Analysis, factor analysis, Z-score, and weight index. Besides, variables used to measure energy poverty including electricity and overall household spending, income per capita, or number of electrical appliances, the table also includes data about socio-demographic factors, including the age, gender, and education level of household heads, number of household members, and geographic information such as urban or rural classification, living area size or housing tenure

Table 4 List of electrical equipment

Non-essential electrical equipment	Essential electrical equipment
Printer	Pumping machine
Fax machine	Sewing machine
Video player, DVD player, digital player, satellite antenna	Color TV
Music rack of various kinds	Black and white TV
Disk player	Radio/radio-cassette player
Computer	Refrigerator
Camera, video recorder	Electric fan
Air conditioner	Electric cooker, electric rice cooker, pressure cooker
Washing machine, clothes-drying machine	
Bath water heater	
Vacuum cleaner, dehumidifier, water filter	
Microwave oven, baking oven	
Juice extractor, citrus juicer	
Other: e.g., Piano, keyboard	

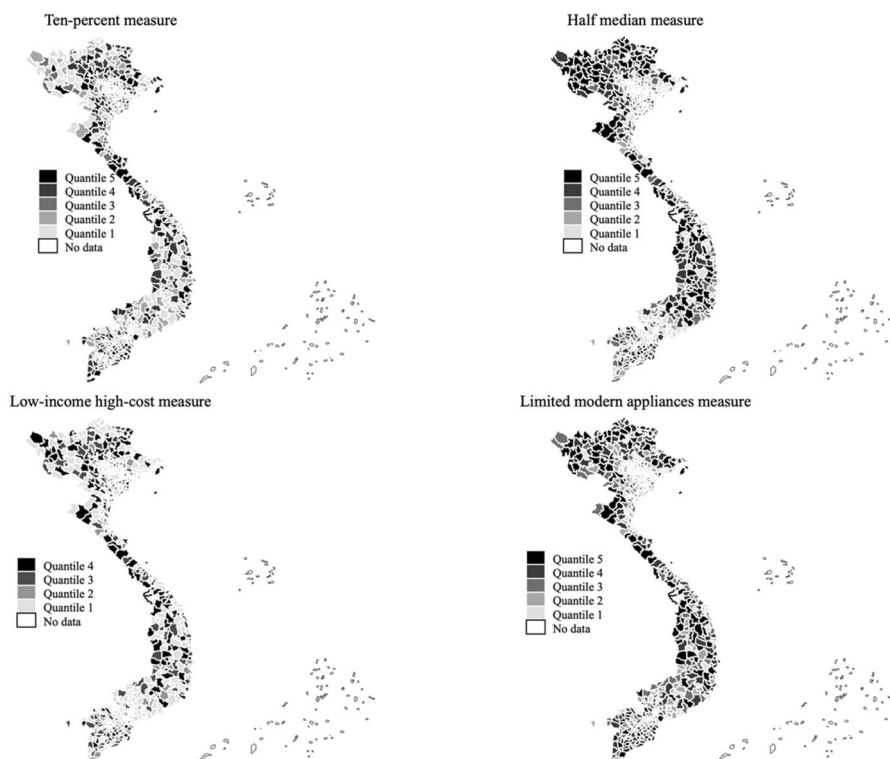


Fig. 10 Different energy poverty individual indices by quintiles. This figure presents maps illustrating the distribution of energy poverty across Vietnam using four individual indices: TPR, M2, LIHC, and MOT for different dimensions of energy poverty. Each map is segmented into quintiles, with darker shades (Quintile 5) representing higher levels of energy poverty and lighter shades (Quintile 1) representing lower levels

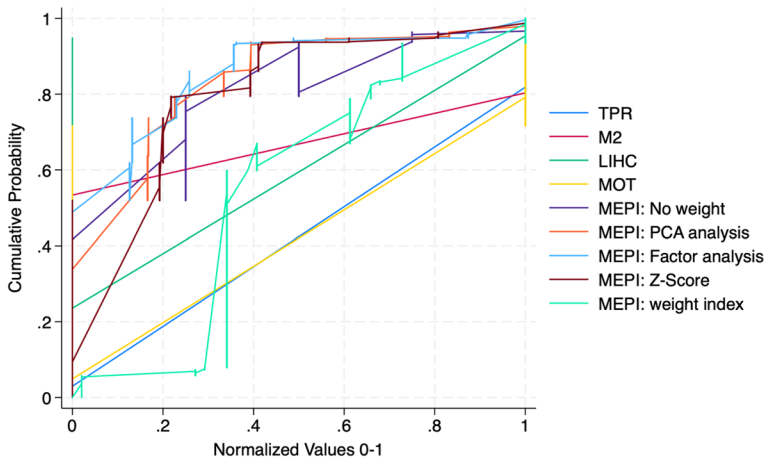


Fig. 11 The cumulative distribution of all energy poverty indices. This figure shows the cumulative distribution of various energy poverty indices (TPR, M2, LIHC, MOT, and MEPI) calculated using different methodologies (unweighted, PCA, FA, Z-score, and weighting by electricity spending). The x-axis represents normalized values ranging from 0 to 1, while the vertical axis shows the cumulative probability

Acknowledgements Authors express our appreciation to helpful comments and suggestions from Editor and two anonymous reviewers of Quality and Quantity journal. All remaining errors are own.

Funding The study is funded the Vietnam Ministry of Education and Training following the Decision 1725/QĐ-BGDĐT dated 26/06/2024 by Minister of the Vietnam Ministry of Education and Training, project number: B2025-KSA-02.

Declarations

Conflict of interest None.

Declaration of generative AI in scientific writing We do not use generative AI in scientific writing upon submission of the paper.

References

- Action, P. (2013). *Poor people's energy outlook 2012: Energy for earning a living*: Practical Action Publishing.
- Ahmed, A., Gasparatos, A.: Multi-dimensional energy poverty patterns around industrial crop projects in Ghana: Enhancing the energy poverty alleviation potential of rural development strategies. *Energy Policy* **137**, 111123 (2020). <https://doi.org/10.1016/j.enpol.2019.111123>
- Al Kez, D., Foley, A., Lowans, C., Del Rio, D.F.: Energy poverty assessment: Indicators and implications for developing and developed countries. *Energy Convers. Manage.* **307**, 118324 (2024). <https://doi.org/10.1016/j.enconman.2024.118324>
- Amin, A., Liu, Y., Yu, J., Chandio, A.A., Rasool, S.F., Luo, J., et al.: How does energy poverty affect economic development? A panel data analysis of South Asian countries. *Environ. Sci. Pollut. Res. Pollut. Res.* **27**(25), 31623–31635 (2020). <https://doi.org/10.1007/s11356-020-09173-6>
- Andrew, S., Romilly, G.: Turning the lights on Sustainable energy and development in Viet Nam. *Overseas Development Institute Case Study Report: Environment*. Overseas Development Institute, London, UK (2014)

- Apergis, N., Polemis, M., Soursou, S.-E.: Energy poverty and education: fresh evidence from a panel of developing countries. *Energy Econ.* **106**, 105430 (2022). <https://doi.org/10.1016/j.eneco.2021.105430>
- Aristondo, O., Onaindia, E.: Inequality of energy poverty between groups in Spain. *Energy* **153**, 431–442 (2018). <https://doi.org/10.1016/j.energy.2018.04.029>
- Astuti, S.P., Day, R., Emery, S.B.: A successful fuel transition? Regulatory instruments, markets, and social acceptance in the adoption of modern LPG cooking devices in Indonesia. *Energy Res. Soc. Sci.* **58**, 101248 (2019). <https://doi.org/10.1016/j.erss.2019.101248>
- Awan, R.U., Sher, F., Abbas, A.: An investigation of multidimensional energy poverty in Pakistan. *Pak. Dev. Rev.* **52**(4), 405–418 (2013)
- Awaworyi Churchill, S., Smyth, R., Trinh, T.-A.: Energy poverty, temperature and climate change. *Energy Econ.* **114**, 106306 (2022). <https://doi.org/10.1016/j.eneco.2022.106306>
- Beresford, M.: Doi Moi in review: the challenges of building market socialism in Vietnam. *J. Contemp. Asia* **38**(2), 221–243 (2008). <https://doi.org/10.1080/00472330701822314>
- Bhatia, M., & Angelou, N. (2014). Capturing the multi-dimensionality of energy access. *Live wire knowledge note series* (Vol. no. 2014/16). Washington, D.C., US: World Bank Group.
- Bhatia, M., & Angelou, N. (2015). *Beyond connections: energy access redefined*: World Bank.
- Boardman, B.: Fuel poverty: from cold homes to affordable warmth. Belhaven Press, London, UK (1991)
- Boivin, J., Ng, S.: Are more data always better for factor analysis? *J. Economet.* **132**(1), 169–194 (2006). <https://doi.org/10.1016/j.jeconom.2005.01.027>
- Campagna, L., Radaelli, L., Ricci, M., Rancilio, G.: Exploring the complexity of energy poverty in the EU: measure it, map it take actions. *Current Sustain/renew. Energy Rep.* **11**(4), 116–126 (2024). <https://doi.org/10.1007/s40518-024-00240-x>
- Castaño-Rosa, R., Sherriff, G., Thomson, H., Guzmán, J.S., Marrero, M.: Transferring the index of vulnerable homes: application at the local-scale in England to assess fuel poverty vulnerability. *Energy Build.* **203**, 109458 (2019a). <https://doi.org/10.1016/j.enbuild.2019.109458>
- Castaño-Rosa, R., Solís-Guzmán, J., Rubio-Bellido, C., Marrero, M.: Towards a multiple-indicator approach to energy poverty in the European Union: A review. *Energy Build.* **193**, 36–48 (2019b). <https://doi.org/10.1016/j.enbuild.2019.03.039>
- Choumert-Nkolo, J., le Roux, L.: Leaving the hearth you know: Internal migration and energy poverty. *World Dev.* **180**, 106628 (2024). <https://doi.org/10.1016/j.worlddev.2024.106628>
- Coelho, S.T., Goldemberg, J.: Energy access: Lessons learned in Brazil and perspectives for replication in other developing countries. *Energy Policy* **61**, 1088–1096 (2013). <https://doi.org/10.1016/j.enpol.2013.05.062>
- Diez, J.R.: Vietnam 30 years after Moi: achievements and challenges. *Zeitschrift Für Wirtschaftsgeographie* **60**(3), 121–133 (2016). <https://doi.org/10.1515/zfw-2016-0035>
- Esquivel García, C.L., Toro-García, G.L.: Multidimensional energy poverty in Colombia: a department-level review from 2018 to 2022. *Heliyon* (2024). <https://doi.org/10.1016/j.heliyon.2024.e34395>
- Feeny, S., Trinh, T.-A., Zhu, A.: Temperature shocks and energy poverty: findings from Vietnam. *Energy Econ.* **99**, 105310 (2021). <https://doi.org/10.1016/j.eneco.2021.105310>
- Galvin, R.: Reducing poverty in the UK to mitigate energy poverty by the 10% and LIHC indicators: What tax changes are needed, and what are the consequences for CO2 emissions? *Ecol. Econ.* **217**, 108055 (2024). <https://doi.org/10.1016/j.ecolecon.2023.108055>
- García Ochoa, R., Graizbord Ed, B.: Privation of energy services in Mexican households: an alternative measure of energy poverty. *Energy Res. Soc. Sci.* **18**, 36–49 (2016). <https://doi.org/10.1016/j.erss.2016.04.014>
- Gencer, D., Meier, P., Spencer, R., & Van, H. T. (2011). State and people, central and local, working together: The Vietnam rural electrification experience. *Asia Sustainable and Alternative Energy Program*. Washington, DC, US: The International Bank for Reconstruction and Development/The World Bank.
- González-Eguino, M.: Energy poverty: An overview. *Renew. Sustain. Energy Rev.* **47**, 377–385 (2015). <https://doi.org/10.1016/j.rser.2015.03.013>
- Guevara, Z., Mendoza-Tinoco, D., Silva, D.: The theoretical peculiarities of energy poverty research: A systematic literature review. *Energy Res. Soc. Sci.* **105**, 103274 (2023). <https://doi.org/10.1016/j.erss.2023.103274>
- Gupta, S., Gupta, E., Sarangi, G.K.: household energy poverty index for India: An analysis of inter-state differences. *Energy Policy* **144**, 111592 (2020). <https://doi.org/10.1016/j.enpol.2020.111592>
- Ha-Duong, M., Nguyen, H.-S.: Subjective satisfaction and objective electricity poverty reduction in Vietnam, 2008–2018. *Fulbright Rev Econ Policy* **1**(1), 43–60 (2021). <https://doi.org/10.1108/FREP-03-2021-0022>

- Ha, N.M., Nguyen, D.L., Trung-Kien, P.: The impact of urbanization on poverty reduction: An evidence from Vietnam. *Cogent Econ Finance* **9**(1), 1918838 (2021). <https://doi.org/10.1080/23322039.2021.1918838>
- Herrero, S.T.: Energy poverty indicators: A critical review of methods. *Indoor Built Environ* **26**(7), 1018–1031 (2017). <https://doi.org/10.1177/1420326x17718054>
- Hills, J. (2011). Fuel poverty: the problem and its measurement. *Case Report*, 69(192).
- Hills, J. (2012). Getting the measure of fuel poverty: Final Report of the Fuel Poverty Review. *Case Report*, 72.
- Jayasinghe, M., Selvanathan, E.A., Selvanathan, S.: Energy poverty in Sri Lanka. *Energy Econ.* **101**, 105450 (2021). <https://doi.org/10.1016/j.eneco.2021.105450>
- Jolliffe, I. T. (2002). Principal Component Analysis and Factor Analysis. In I. T. Jolliffe (Ed.), *Principal Component Analysis* (pp. 150–166). New York, NY: Springer New York.
- Kashour, M., Jaber, M.M.: Revisiting energy poverty measurement for the European Union. *Energy Res. Soc. Sci.* **109**, 103420 (2024). <https://doi.org/10.1016/j.erss.2024.103420>
- Katoch, O.R., Sharma, R., Parihar, S., Nawaz, A.: Energy poverty and its impacts on health and education: a systematic review. *Int. J. Energy Sect. Manage.* **18**(2), 411–431 (2024). <https://doi.org/10.1108/IJESM-10-2022-0007>
- Kooijman-van Dijk, A.L., Clancy, J.: Impacts of electricity access to rural enterprises in Bolivia, Tanzania and Vietnam. *Energy Sustain. Dev.* **14**(1), 14–21 (2010). <https://doi.org/10.1016/j.esd.2009.12.004>
- Le, V.T., Pitts, A.: A survey on electrical appliance use and energy consumption in Vietnamese households: Case study of Tuy Hoa city. *Energy Build.* **197**, 229–241 (2019). <https://doi.org/10.1016/j.enbuild.2019.05.051>
- Legendre, B., Ricci, O.: Measuring fuel poverty in France: Which households are the most fuel vulnerable? *Energy Econ.* **49**, 620–628 (2015). <https://doi.org/10.1016/j.eneco.2015.01.022>
- Li, X., Smyth, R., Xin, G., Yao, Y.: Warmer temperatures and energy poverty: Evidence from Chinese households. *Energy Econ.* **120**, 106575 (2023). <https://doi.org/10.1016/j.eneco.2023.106575>
- Liddell, C., Morris, C., McKenzie, S.J.P., Rae, G.: Measuring and monitoring fuel poverty in the UK: National and regional perspectives. *Energy Policy* **49**, 27–32 (2012). <https://doi.org/10.1016/j.enpol.2012.02.029>
- Lin, B., Wang, Y.: Does energy poverty really exist in China? From the perspective of residential electricity consumption. *Energy Policy* **143**, 111557 (2020). <https://doi.org/10.1016/j.enpol.2020.111557>
- Lu, S., Ren, J.: A comprehensive review on energy poverty: definition, measurement, socioeconomic impact and its alleviation for carbon neutrality. *Environ. Dev. Sustain.* (2023). <https://doi.org/10.1007/s10668-023-04143-7>
- Lukyanets, A.S., Nguen, T.K., Ryazantsev, S.V., Tikunov, V.S., Pham, H.H.: Influence of climatic changes on population migration in Vietnam. *Geogr. Nat. Resour. Nat. Resour.* **36**(3), 313–317 (2015). <https://doi.org/10.1134/S1875372815030129>
- Malesky, E. (2022). Decentralization and Economic Development in Vietnam. In *Routledge Handbook of Contemporary Vietnam* (pp. 203–219): Routledge.
- Manasi, B., Mukhopadhyay, J.P.: Definition, measurement and determinants of energy poverty: Empirical evidence from Indian households. *Energy Sustain. Dev.* **79**, 101383 (2024). <https://doi.org/10.1016/j.esd.2024.101383>
- Maruejols, L., Höschle, L., Yu, X.: Vietnam between economic growth and ethnic divergence: A LASSO examination of income-mediated energy consumption. *Energy Econ.* **114**, 106222 (2022). <https://doi.org/10.1016/j.eneco.2022.106222>
- März, S.: Assessing the fuel poverty vulnerability of urban neighbourhoods using a spatial multi-criteria decision analysis for the German city of Oberhausen. *Renew. Sustain. Energy Rev.* **82**, 1701–1711 (2018). <https://doi.org/10.1016/j.rser.2017.07.006>
- McCaig, B., McMillan, M. S., Ngoma, M. M., & Pham, A. (2024). Industrial Zones and Private Formal Sector Job Creation in Vietnam.
- Middlemiss, L., Ambrosio-Albalá, P., Emmel, N., Gillard, R., Gilbertson, J., Hargreaves, T., et al.: Energy poverty and social relations: A capabilities approach. *Energy Res. Soc. Sci.* **55**, 227–235 (2019). <https://doi.org/10.1016/j.erss.2019.05.002>
- Min, B., O’Keeffe, Z.P., Abidoye, B., Gaba, K.M., Monroe, T., Stewart, B.P., et al.: Lost in the dark: A survey of energy poverty from space. *Joule* **8**(7), 1982–1998 (2024). <https://doi.org/10.1016/j.joule.2024.05.001>
- Minh, H.-D., & Son, N. H. Electricity poverty reduction and progress towards the sustainable development goal 7: Vietnam, 2008–2018. In *1st IAEE Online Conference, Online conference, 2021: The International Association for Energy Economics*

- Moore, R.: Definitions of fuel poverty: Implications for policy. *Energy Policy* **49**, 19–26 (2012). <https://doi.org/10.1016/j.enpol.2012.01.057>
- Moss, T., Bazilian, M., Blimpo, M., Culver, L., Kincer, J., Mahadavan, M., et al. (2020). The Modern Energy Minimum: The case for a new global electricity consumption threshold. *Energy for Growth Hub*, 30.
- Nguyen, C.P., Dinh Thanh, S.: Nexus between energy poverty and economic vulnerability: Evidence from low and middle income economies. *Energy Sour. Part B* **17**(1), 2056264 (2022). <https://doi.org/10.1080/15567249.2022.2056264>
- Nguyen, C.P., Nasir, M.A.: An inquiry into the nexus between energy poverty and income inequality in the light of global evidence. *Energy Econ.* **99**, 105289 (2021). <https://doi.org/10.1016/j.eneco.2021.105289>
- Nguyen, C.P., Su, T.D.: Alleviating energy poverty for forest conservation: it seems to work, but what are we missing? *Land Use Policy* **109**, 105625 (2021a). <https://doi.org/10.1016/j.landusepol.2021.105625>
- Nguyen, C.P., Su, T.D.: Does energy poverty matter for gender inequality? Global evidence. *Energy Sustain. Dev.* **64**, 35–45 (2021b). <https://doi.org/10.1016/j.esd.2021.07.003>
- Nguyen, C.P., Su, T.D.: The ‘vicious cycle’ of energy poverty and productivity: insights from 45 developing countries. *Environ. Sci. Pollut. Res. pollut. Res.* **28**(40), 56345–56362 (2021c). <https://doi.org/10.1007/s11356-021-14614-x>
- Nguyen, C.P., Su, T.D.: The influences of government spending on energy poverty: evidence from developing countries. *Energy* **238**, 121785 (2022). <https://doi.org/10.1016/j.energy.2021.121785>
- Nguyen, L.T.: Vietnamese energy sector: Challenges and opportunities. Griffith University, Australia (2022)
- Nguyen, T. H. T., & Nguyen-Ngoc, S. (2025). Factors Affecting Consumers’ Purchase Intention Toward Energy-efficient Household Appliances in Ho Chi Minh City – Vietnam. In J. Paliszkievicz, D. Varoğlu, & O. Kulykovets (Eds.), *Future Workscapes: Strategic Insights and Innovations in Human Resources and Organizational Development* (Vol. 11A, pp. 63–79, International Perspectives on Equality, Diversity and Inclusion): Emerald Publishing Limited.
- Nguyen, T.T., Nguyen, T.-T., Hoang, V.-N., Wilson, C., Managi, S.: Energy transition, poverty and inequality in Vietnam. *Energy Policy* **132**, 536–548 (2019). <https://doi.org/10.1016/j.enpol.2019.06.001>
- Nussbaumer, P., Bazilian, M., Modi, V.: Measuring energy poverty: Focusing on what matters. *Renew. Sustain. Energy Rev.* **16**(1), 231–243 (2012). <https://doi.org/10.1016/j.rser.2011.07.150>
- Nussbaumer, P., Nerini, F.F., Onyeji, I., Howells, M.: Global insights based on the multidimensional energy poverty index (MEPI). *Sustainability* **5**(5), 2060–2076 (2013)
- Okushima, S.: Gauging energy poverty: A multidimensional approach. *Energy* **137**, 1159–1166 (2017). <https://doi.org/10.1016/j.energy.2017.05.137>
- Olang, T.A., Esteban, M., Gasparatos, A.: Lighting and cooking fuel choices of households in Kisumu City, Kenya: A multidimensional energy poverty perspective. *Energy Sustain. Dev.* **42**, 1–13 (2018). <https://doi.org/10.1016/j.esd.2017.09.006>
- Olawumi Israel-Akinbo, S., Snowball, J., Fraser, G.: An investigation of multidimensional energy poverty among south african low-income households. *South African J. Econ.* **86**(4), 468–487 (2018). <https://doi.org/10.1111/saje.12207>
- Omar, M.A., Hasanujjaman, M.: Multidimensional energy poverty in Bangladesh and its effect on health and education: a multilevel analysis based on household survey data. *Energy Policy* **158**, 112579 (2021). <https://doi.org/10.1016/j.enpol.2021.112579>
- Pachauri, S., Spreng, D.: Measuring and monitoring energy poverty. *Energy Policy* **39**(12), 7497–7504 (2011). <https://doi.org/10.1016/j.enpol.2011.07.008>
- Pérez-Fargallo, A., Marín-Restrepo, L., Contreras-Espinoza, S., Bienvenido-Huertas, D.: Do we need complex and multidimensional indicators to assess energy poverty? The case of the Chilean indicator. *Energy Build.* **295**, 113314 (2023). <https://doi.org/10.1016/j.enbuild.2023.113314>
- Pouran, H., Lopes, M.P., Ziar, H., Branco, D.A., Sheng, Y.: Evaluating floating photovoltaics (FPVs) potential in providing clean energy and supporting agricultural growth in Vietnam. *Renew. Sustain. Energy Rev.* **169**, 112925 (2022). <https://doi.org/10.1016/j.rser.2022.112925>
- Que, N.D., Van Song, N., Thuan, T.D., Van Tien, D., Van Ha, T., Phuong, N.T.M., et al.: How temperature shocks impact energy poverty in Vietnam: mediating role of financial development and environmental consideration. *Environ. Sci. Pollut. Res. pollut. Res.* **29**(37), 56114–56127 (2022). <https://doi.org/10.1007/s11356-022-19672-3>
- Reddy, A. K., Annecke, W., Blok, K., Bloom, D., Boardman, B., Eberhard, A., et al. (2000, p.44). Energy and social issues. In WorldEnergyCouncil, & UNEP (Eds.), *World energy assessment: Energy and the challenge of sustainability* (pp. 39–60). NewYork, US.

- Ringnér, M.: What is principal component analysis? *Nat. Biotechnol.* **26**(3), 303–304 (2008). <https://doi.org/10.1038/nbt0308-303>
- Rizal, R.N., Hartono, D., Dartanto, T., Gultom, Y.M.L.: Multidimensional energy poverty: a study of its measurement, decomposition, and determinants in Indonesia. *Heliyon* (2024). <https://doi.org/10.1016/j.heliyon.2024.e24135>
- Sadath, A.C., Acharya, R.H.: Assessing the extent and intensity of energy poverty using multidimensional energy poverty index: Empirical evidence from households in India. *Energy Policy* **102**, 540–550 (2017). <https://doi.org/10.1016/j.enpol.2016.12.056>
- Salman, M., Zha, D., Wang, G.: Assessment of energy poverty convergence: A global analysis. *Energy* **255**, 124579 (2022). <https://doi.org/10.1016/j.energy.2022.124579>
- Siksnelyte-Butkiene, I., Streimikiene, D., Lekavicius, V., Balezentis, T.: Energy poverty indicators: A systematic literature review and comprehensive analysis of integrity. *Sustain. Cities Soc.* **67**, 102756 (2021). <https://doi.org/10.1016/j.scs.2021.102756>
- Simcock, N. (2020, p.1). Energy. In A. Kobayashi (Ed.), *International Encyclopedia of Human Geography (Second Edition)* (pp. 123–135). Oxford: Elsevier.
- Solaymani, S., Vaghefi, N., Kari, F.: The multidimensional poverty measure among Malaysian employee provident fund (EPF) retirees. *Appl. Res. Qual. Life* **14**(5), 1353–1371 (2019). <https://doi.org/10.1007/s11482-018-9658-4>
- Son, H., Yoon, S.: Reducing energy poverty: characteristics of household electricity use in Vietnam. *Energy Sustain. Dev.* **59**, 62–70 (2020). <https://doi.org/10.1016/j.esd.2020.08.007>
- Sovacool, B.K.: The political economy of energy poverty: A review of key challenges. *Energy Sustain. Dev.* **16**(3), 272–282 (2012). <https://doi.org/10.1016/j.esd.2012.05.006>
- Sy, S.A., Mokaddem, L.: Energy poverty in developing countries: a review of the concept and its measurements. *Energy Res. Soc. Sci.* **89**, 102562 (2022). <https://doi.org/10.1016/j.erss.2022.102562>
- Trinh, T.T., Munro, A.: Climate change and migration decisions: a choice experiment from the Mekong Delta, Vietnam. *Ecol. Econ.* **224**, 108307 (2024). <https://doi.org/10.1016/j.ecolecon.2024.108307>
- Tripathi, A., Sagar, A.D., Smith, K.R.: Promoting clean and affordable cooking: smarter subsidies for LPG. *Econ. Pol. Wklywky* **50**(48), 81–84 (2015)
- UNDP (2007). Energy and poverty in Vietnam: Challenges and the Way forward. In V. J. K. Marcia (Ed.), *Regional Energy Programme for Poverty Reduction UNDP Regional Centre in Bangkok*. Bangkok, Thailand: United Nations Development Programme.
- Vo, T.T., Nguyen, D.X.: Impact of trade liberalization on household welfare: an analysis using household exposure-to-trade indices. *Soc. Indic. Res.* **153**(2), 503–531 (2021). <https://doi.org/10.1007/s11205-020-02499-1>
- Wang, F., Geng, H., Zha, D., Zhang, C.: Multidimensional energy poverty in China: measurement and spatio-temporal disparities characteristics. *Soc. Indic. Res.* **168**(1), 45–78 (2023). <https://doi.org/10.1007/s11205-023-03129-2>
- Zhang, D., Li, J., Han, P.: A multidimensional measure of energy poverty in China and its impacts on health: An empirical study based on the China family panel studies. *Energy Policy* **131**, 72–81 (2019). <https://doi.org/10.1016/j.enpol.2019.04.037>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Nguyen Doan¹ · Canh Phuc Nguyen^{1,2}  · Thanh Dinh Su¹ · Binh Quang Nguyen¹ · Duyen Thuy Le Tran³

✉ Canh Phuc Nguyen
canhnguyen@ueh.edu.vn

Nguyen Doan
nguyen.doan@ueh.edu.vn

Thanh Dinh Su
dinhthanh@ueh.edu.vn

Binh Quang Nguyen
binhnq@ueh.edu.vn

Duyen Thuy Le Tran
duyentlt@ueh.edu.vn

¹ School of Public Finance, College of Economics, Law and Government, University of Economics Ho Chi Minh City, 59C Nguyen Dinh Chieu, District 3, Ho Chi Minh City 700000, Vietnam

² Health and Agricultural Policy Research Institute, College of Economics, Law and Government, University of Economics Ho Chi Minh City, 279 Nguyen Tri Phuong, District 10, Ho Chi Minh City 700000, Vietnam

³ University of Economics Ho Chi Minh City, 59C Nguyen Dinh Chieu, District 3, Ho Chi Minh City 700000, Vietnam